Values, Temporality, and AI use when Negotiating High-Subjectivity Medical Decision-Making

Introduction

Healthcare decision-making often involves high subjectivity decisions, where clinical evidence alone cannot determine the best course of action. In such cases, the patient's individual values—preferences, beliefs, and priorities—play a central role in negotiating trade-offs [1,4] . This is especially evident in complex care scenarios, such as whether to initiate or withhold interventions like gastric tube feeding for people with advanced dementia.

LLMs within AI-enhanced systems offer the potential to store, retrieve, and interpret expressed values to support decision-making. Doing so may help reduce the current high cost of elicitation - requiring time-intensive, human-present, synchronous service provision, discourages repeated elicitation.

As healthcare systems increasingly explore AI-enhanced tools to support decision-making, a critical challenge of temporality emerges: how to represent, interpret, and update human values over time. This position paper references increasing evidence that patient values are dynamic rather than static, and that any AI system designed to incorporate them must account for their **temporality**.

1. The Role of Values in High-Subjectivity Decisions

Certain healthcare decisions resist purely objective resolution. While clinical data can inform prognosis and treatment options, they cannot determine what constitutes an acceptable quality of life or a meaningful outcome for the patient [7].

For example, decisions about gastric tube feeding in patients with advanced dementia often hinge on value-laden considerations: prioritization of longer life versus comfort, the meaning and relative value of dignity, and personal motivations to undergo suffering. The decision is complicated when different decision stakeholders—patients (when previously expressed), family members, and clinicians—may hold divergent views [5], thereby requiring an active negotiation of shared or divergent values.

Thus, may assist or complicate the effort to identify the appropriate path forward, or lead. The presence of shared or divergent values have to be identified, prioritized and moved toward unified understandings. The inclusion of decision-support systems adds a fourth voice that may or may not represent the values of their owners well. It follows then, that any decision-support system that fails to account for and represent the owner's values risks producing recommendations that are technically sound but potentially misaligned with the pursuit of quality of life.

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2. AI-Enhanced Systems and the Temporality of Personal Values

Most current approaches in healthcare implicitly treat values as static inputs—captured once and applied indefinitely. This assumption is flawed. Instead, research in HCI suggests that values evolve over time [2,3] possibly influenced by factors such as changes in health status, life experiences and shifting priorities, aging and developmental stages, social and cultural influences. In other words, a preference expressed at age 40 may not accurately reflect the priorities of the same individual at age 75. Similarly, a person's tolerance for disability or dependence may change after experiencing illness firsthand—a phenomenon related to response shift and adaptation [6,8]. In summary there is evidence that values *evolve*.

Therefore, moving forward with human values in AI-enhanced decision support systems should incorporate a temporal dimension of values, recognizing that elicited values have a "timestamp" and a context of elicitation. Consequentially, systems that use "older" values may be using values that have decayed in relevance, creating conflicts between past and present values. In other words, rather than treating values as *fixed* directives, systems may need to conceptualize values as *evolving trajectories*.

3. Elicitation and Updating of Evolving Values

As indicated above, the conception of what is "older" requires inquiry. How and when values should be elicited and updated is a key design question. Preliminary ideas include 1) routinized elicitation (e.g. during checkups), 2) at major life transitions (e.g. retirement or diagnosis) and more obviously 3) when individuals engage in care planning. Clearly, early elicitation might allow for reflection without the pressure of imminent decisions, but these elicited values must be understood as provisional rather than definitive. However, as with many interactive systems that seek life-cogent input from humans, excessive prompting may lead to fatigue or disengagement.

This leads to a research agenda that includes balancing responsiveness with respect for user burden. Alternatively, the potential for user burden creates an argument for automated values updating, based on known population level factors linked to aging, health status changes or population data for example.

While the latter may offer efficiency, it raises further significant concerns that bear exploration and explication. For example, it risks stereotyping individuals based on demographic averages, may override authentic, previously expressed preferences and it lacks transparency thereby undermining trust. While intriguing as a research topic, automated updating should be approached with caution and supporting evidence.

Hence, a research agenda for AI-enhanced systems in high-subjectivity decision making might next support the values elicitation process by accompanying users through the steps of values evolution - identifying when updates may be needed, presenting prior values alongside current context, highlighting potential inconsistencies and guiding insight into personal values.

Conclusion

Incorporating values into healthcare decision-making is both necessary and complex. As AI systems become more integrated into healthcare, we should recognize that values evolve, move beyond static representations of human preferences, and instead engage with the temporal, evolving nature of values.

Designing such systems requires careful attention to the contextual circumstances of values elicitation, evidence for programmatic updating that balances undue burden with assurance of relevance, and the role of human agency throughout. Ultimately, the goal is to better support individuals and clinicians in making decisions that remain aligned with what matters most over time.

References

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